

WIDIANA ANGGREANI RAHWAL

<https://widiaangrn.netlify.app> | widia.angrn@gmail.com | www.linkedin.com/in/widiaangrn

SUMMARY

Software developer with experience building web and mobile applications across multiple industries, including oil & gas, healthcare, education, tourism, and retail. Proficient in ASP.NET MVC, Django, Flutter, and Laravel for developing industrial monitoring systems, operational dashboards, automation systems, and transaction applications. Skilled in user requirements analysis, database performance optimization, and building scalable, maintainable, high-performance applications.

WORK EXPERIENCE

Software Developer

PT. Sila Teknologi Utama

January 2024 – Present

Key Achievements:

- Developed 19 functional requirements (9 dashboards & toolboxes, 10 other features) for Pertamina Hulu Mahakam's oil & gas well monitoring system using ASP.NET MVC and MSSQL / Oracle SQL.
- Revamped application performance by migrating business logic and calculations from the application layer to Stored Procedures and SQL Views with parameter filtering and pagination, reducing data transferred per request from millions of records to 50 records — significantly improving access speed.
- Contributed to the development of more than 10 applications across multiple sectors (education, healthcare, retail, and tourism) using Django and Flutter with PostgreSQL and Oracle SQL databases.
- Developed a ticketing platform, multi-method payment system (OTS, QRIS, EDC) integrated with Midtrans, along with voucher and promotion systems for the tourism and retail sectors.
- Optimized backend architecture and system security to improve code scalability and maintainability.

Project Details:

- A. Automation & Industrial Monitoring System Development (Client: Pertamina Hulu Mahakam)
 - Designed and developed interactive custom charts using Chart.js for visualizing gas, oil, and sand risk assessment data, enabling faster issue identification.
 - Designed an Entity-Relationship Diagram (ERD) to consolidate multiple databases into a centralized data warehouse, improving data structure and control.
 - Developed 19 functional requirements using ASP.NET MVC and MSSQL / Oracle SQL, covering 9 dashboards / toolboxes and 10 supporting features such as data tables, Excel upload, and landing pages.
 - Revamped application performance by moving calculation logic from the application layer to Stored Procedures and SQL Views, reducing data volume per request from millions of records to 50 records through database-level parameter filtering and pagination.
- B. Web Feature Development (Django)
 - Optimized authentication systems by implementing OOP principles, Role-Based Access Control (RBAC), and automated activation token delivery via email, improving user registration security.
 - Developed a real-time chat feature using Django Channels, ASGI, and Redis to support stable and responsive two-way communication.
 - Developed payment modules (OTS, QRIS, EDC, etc.) integrated with Midtrans along with voucher and promotion systems.
 - Developed schedulers using django-crontab to automate daily data synchronization and email notification delivery (alerts, transaction confirmations).
 - Redesigned several pages and developed other key features.
 - Developed a patient registration module integrated with queue systems, barcodes, and online forms, reducing registration time and minimizing input errors.
- C. Mobile App Development & Integration (Flutter)
 - Designed UI/UX and developed Flutter applications along with REST APIs to ensure reliable real-time data synchronization for transactions, inventory, pricing, tickets, and other data.
 - Developed medical sample identification through a scannable barcode-based labeling system to instantly display patient data.
- D. E-Learning & Assessment System Development
 - Developed an Exam System from UI/UX design through implementation, covering 4 user roles (Admin, Teacher, Exam Supervisor, Student) with distinct permissions and workflows.
 - Built a real-time exam supervision module, including student activity monitoring, lock/unlock access, and remaining time display.
 - Configured Safe Exam Browser (SEB) to restrict exam access on laptops to a controlled browser environment.

Software Quality Assurance

PT. Mahakarya Inovasi Bangsa

October 2023 – December 2023

- Conducted User Acceptance Testing (UAT) and System Integration Testing (SIT) on a Laravel-based website and compiled test result documentation.

- Performed bug analysis, communicated findings to the development team, and produced comprehensive testing reports using Trello.

EDUCATION & TRAINING

Kalimantan Institute of Technology

2019 – 2023

Bachelor of Informatics, 3.76/4.00

- Developed ITK's Internal Quality Assurance Information System end-to-end using the Personal Extreme Programming methodology, built with Laravel and MySQL to streamline academic accreditation data management and reporting.
- Built various learning projects to master full-stack development using Laravel (backend), React.js (frontend), and MySQL (database).
- Technical: PHP, Laravel, MySQL, React.js, VirtualBox, Docker (Basic).

IDCamp 2024

September 2024 – May 2025

Multi-Platform App Developer

- Proficient in Flutter application development, covering responsive UI/UX design, state management with Provider, and reliable REST API integration.

Google Bangkit 2022

February 2022 – July 2022

Android Developer Learning Path

- Developed DentsHealth, a dental health diagnostic application built with Kotlin, leveraging Natural Language Processing (NLP) to analyze user complaints and cloud computing for Machine Learning model deployment, resulting in automated detection of dental and oral health issues.
- Studied Android development skills including Kotlin, Android fundamentals, UX design, SOLID principles, and OOP paradigms.
- Developed professional soft skills including time management, adaptability, critical thinking, and English communication.

ORGANIZATIONAL EXPERIENCE

Informatics Student Association (HMIF)

September 2020 – February 2021

Media & Information Staff

- Designed Instagram content for HMIF using Adobe Illustrator and CorelDraw, and participated as a committee member in organizational events.

SKILLS

- **Programming language:** Python, Dart, JavaScript (jQuery), PHP, C#
- **Framework:** Django, Flutter, Laravel, ASP.NET MVC
- **Database:** PostgreSQL, MySQL, Oracle SQL
- **Development Tools:** Git, Postman, Docker (Basic), VirtualBox, XAMPP, Laragon
- **Design & Collaboration Tools:** Figma, Draw.io, Miro, Trello
- **Front-End Tools:** React.js (Basic), Bootstrap, HTML, CSS, Chart.js
- **Non-Technical:** Analytical thinking, time management, teamwork & collaboration, adaptability, problem solving.
- **Language:** Indonesian (Native), English (Intermediate | IAET ITK: 520, TOEFL Prediction: 610)

ACHIEVEMENTS & OTHER EXPERIENCES

- **Copyright & Certificate:** Software Copyright for PsyCare, 15+ technology certifications from Dicoding, IDCamp, Google Bangkit, and other programs including Flutter, Android Development, Dart, SOLID Principles, and Fundamental AI.
- **Other Projects:** CISA (Covid-19 Information and Self Check Up Application), PetYourself, PsyCare, SIKMA (Internship Information System), DentsHealth, Restaurant App, SIMJAMU (Quality Assurance System), LeafPad, Gallery App.
- **Volunteer Work:** Mangrove Warrior 2025, Lestravida Balikpapan Batch 1.